

# DrainBot Setup Guide

How to load and run the project on another Windows computer

**GitHub repository:** [https://github.com/sandhiyanambiraj/drain\\_bot](https://github.com/sandhiyanambiraj/drain_bot)

This guide runs the camera vision console. The camera requires a working webcam and browser permission.

## 1. Install required software

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Install these programs on the new computer:

- **Git for Windows:** <https://git-scm.com/download/win>
- **Node.js LTS:** <https://nodejs.org/>
- **Python 3.11 or newer:** <https://www.python.org/downloads/>

During Python installation, enable **Add Python to PATH**.

## 2. Download the project

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Open PowerShell and run:

```
git clone https://github.com/sandhiyanambiraj/drain_bot.git
cd drain_bot
```

## 3. Install the frontend packages

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```
cd frontend
npm install
```

Wait until the installation finishes without errors.

## 4. Start the camera vision console

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From the `frontend` folder, run:

```
npm run dev
```

Open the URL shown in the terminal. It is normally:

```
http://localhost:5173/
```

## 5. Enable the camera

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1. Click **Use my camera**.
2. When the browser asks for permission, choose **Allow**.
3. If permission was blocked, click the camera or lock icon beside the address bar.
4. Set Camera to **Allow**, reload the page, and click **Use my camera** again.
5. Hold a plastic bottle clearly in front of the camera with good lighting.

The status should show `AI model ready`. A recognized bottle is shown as `PLASTIC DETECTED • bottle`.

## 6. Test with a photo

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If camera permission is unavailable:

1. Click **Scan photo**.
2. Select a clear image of a plastic bottle.
3. Read the result in the live status area.

## 7. Optional production preview

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To build and preview the production version:

```
cd frontend  
npm run build  
npm run preview
```

Open the preview URL shown in the terminal, normally `http://localhost:4173/`.

## 8. Common problems

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### “npm is not recognized”

Install Node.js LTS, close PowerShell, open a new PowerShell window, and try again.

### “Camera blocked”

Allow camera access from the browser address-bar permissions, reload the page, and try again.

### “AI model unavailable”

Make sure the computer has internet access, wait for the model to load, or click **Retry**.

### No bottle detected

Use a clear bottle image, improve lighting, keep the bottle fully visible, and move it closer to the camera. The model recognizes common object shapes; it cannot guarantee the material of every object.

### Port already in use

Run `npm run dev -- --port 5174` and open the new URL shown in the terminal.

## 9. Stop the application

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Return to the PowerShell window running the server and press `Ctrl + C`.

DrainBot Vision Console setup guide. Repository: [sandhiyanambiraj/drain\\_bot](https://github.com/sandhiyanambiraj/drain_bot)